SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID - SIH25019**
- **Problem Statement Title - Digital Learning Platform for Rural School Students in Nabha**
- **Theme - Smart Education**
- **PS Category - Software**
- **Team ID - 75659**
- **Team Name - TECH IQ**

TECH IQ

A TRANSFORMING DIGITAL PLATFORM FOR INCLUSIVE RURAL EDUCATION

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❖ Addressing the Problem:

- Rural schools face **poor internet, outdated systems**, and **low digital literacy**.
- Students struggle with accessing quality educational resources.
- Teachers lack tools to track progress effectively.

❖ Innovation and uniqueness

Offline + Multi-Language Access
Learn Anytime, Anywhere

Voice and Dyslexia Friendly
Inclusive for all Learners

❖ Our Solution

Offline Access
Download quizzes, notes, PDFs, videos, and assignments for offline use.

Multi-Language Support
Content available in 8+ Indian languages.

AI-Powered Assistance
Smart tutor guides students and provides interview prep support.

Interactive Learning
Gamified quizzes, coding practice, badges, and leaderboards.

Teacher & Parent Dashboard
Track attendance, assignments, and student progress.

Code Editor
Run programs anytime offline or online.

Voice Commands & Audio Guidance
Accessible for students with reading difficulties.

A single platform for students, teachers, and parents, designed for low connectivity areas.

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TECHNICAL APPROACH

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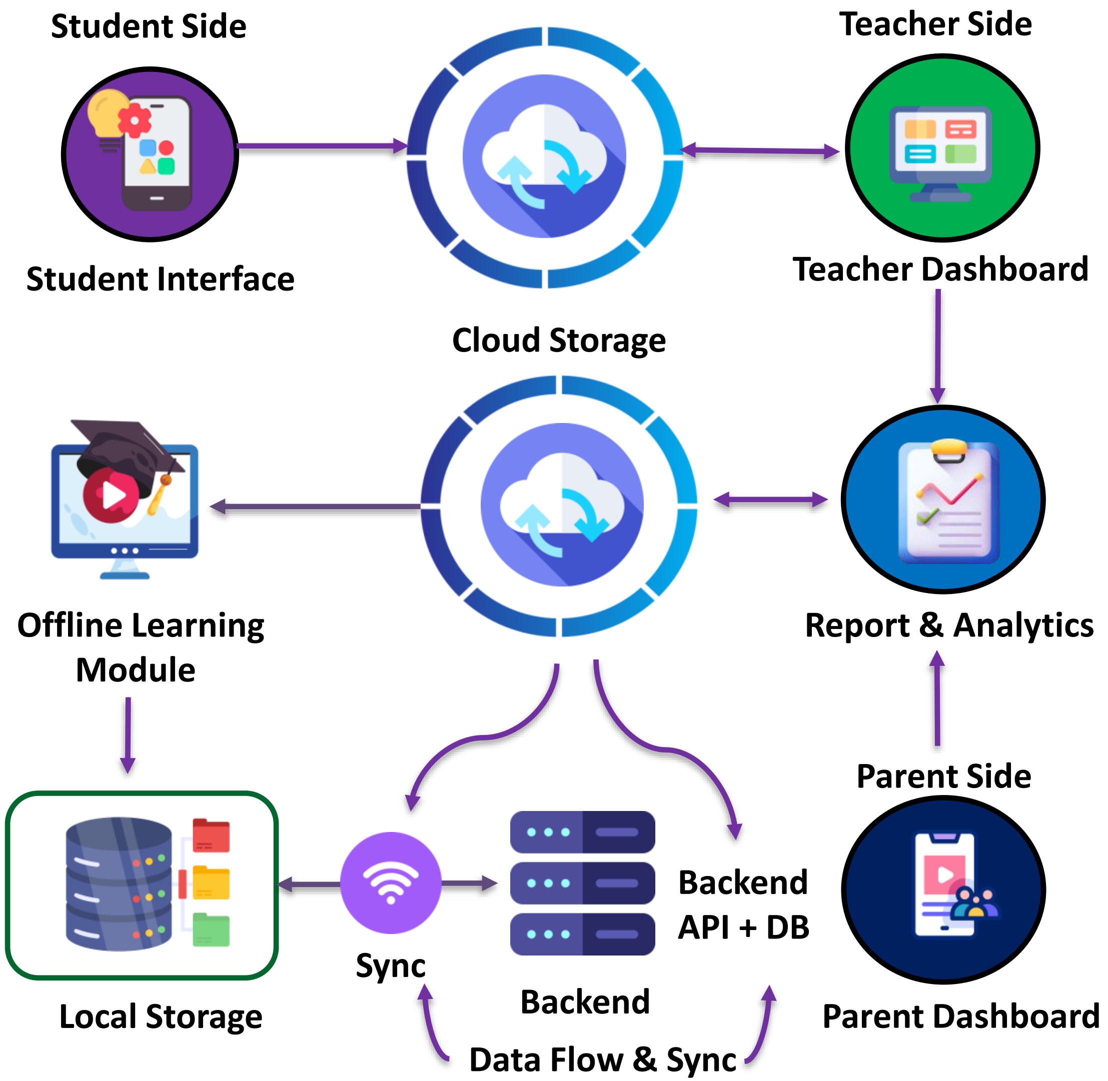
❖ Technology Stack (Web + Mobile)

Component	Technology	Purpose
Mobile App	Flutter (Dart)	Cross-platform app with offline & multi-language support
Web App	Next.js (React) + Tailwind CSS	Responsive interface for students, teachers, parents
Backend	Django + Django REST Framework	APIs for users, content, quizzes, attendance
Database (Online)	Firebase Firestore	Real-time data sync, progress, leaderboards
Database (Offline)	SQLite (mobile), IndexedDB (web)	Local storage & offline access
AI Tutor	Python ML + TensorFlow / Gemini AI API + Grok AI	Personalized learning, career & interview guidance
Voice & Accessibility	TTS + STT (Vosk, Whisper, Google)	Voice commands, audio mode, inclusivity
Peer Learning	Bluetooth / WiFi Hotspot	Offline content sharing between devices
Community Mode	Local storage + Hotspot Sync	Emergency updates & village notice board
Cloud Storage	Firebase Storage / Cloudinary	Store PDFs, notes, videos, media



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FEASIBILITY AND VIABILITY

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Feasibility

1. Works on low-cost smartphones and tablets.
2. Can be used **without internet** and syncs later when online.
3. Easy to use for students, teachers, and parents.
4. Built on reliable and widely used technologies.

Viability

1. Low setup and running cost for schools.
2. Fits rural areas where internet and devices are limited.
3. Can scale to more schools and regions in the future.
4. Parents and teachers need very little training.

Challenges

1. Some students may not have their own devices.
2. Parents might struggle to use apps.
3. Power outages and weak internet in rural areas
4. Keeping students interested for long-term use..

Solutions

1. Provide shared devices at schools or community centers.
2. Conduct easy-to-follow training sessions for parents.
3. Include fun, reward-based learning to keep students engaged.
4. Make the app fully usable offline.

Supporting Facts for Feasibility and Viability

→ **Offline-first apps** help students continue learning and improve retention by up to 70%, even in areas with poor internet like rural Nabha.

→ **Easy-to-use** design in local languages helps students and teachers use the platform comfortably and regularly.

→ **Digital platforms** give rural students better access to quality learning, reduce the rural-urban gap, and build future digital skills.

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IMPACT AND BENEFITS

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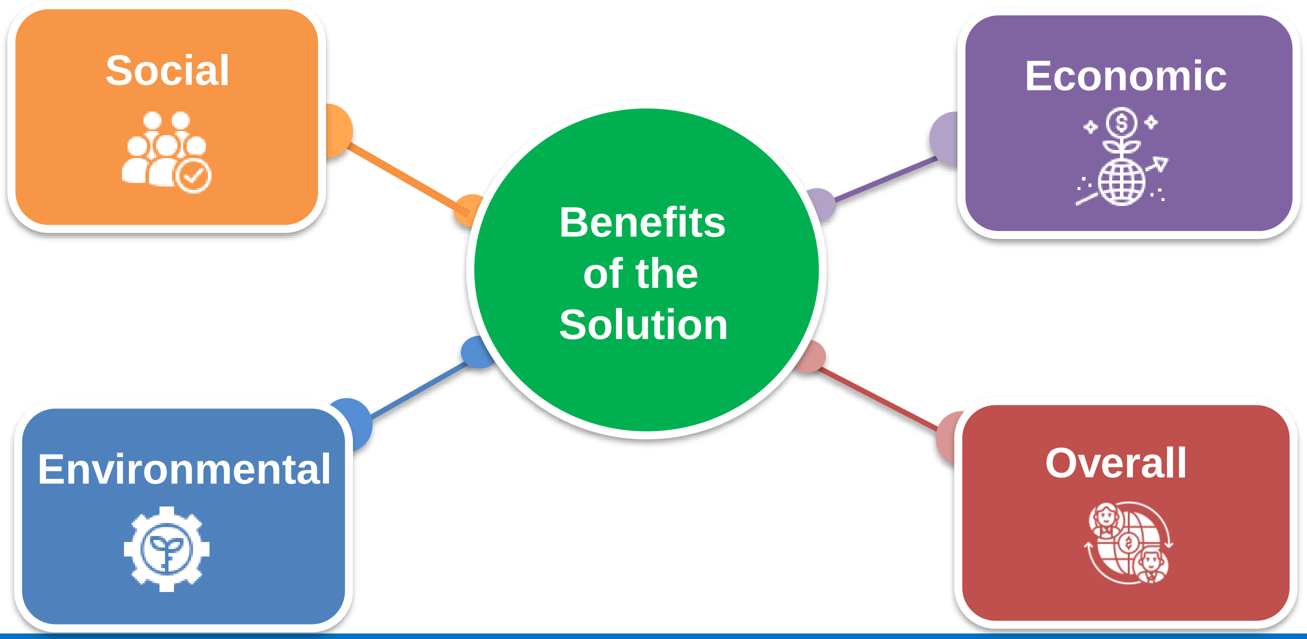
❖ Potential Impact on the Target Audience

- Students in rural Nabha will get access to quality learning materials even without internet.
- Teachers will be able to track student progress and provide personalized support.
- Schools will be able to bridge the gap between rural and urban education standards.
- Rural students will get equal exposure to modern digital tools as urban students.



❖ Benefits of the Solution

Social

- Provides equal learning opportunities for all students.
- Encourages inclusive education and helps students with limited resources.

Economic

- Reduces the cost of printed textbooks and private coaching.
- Prepares students for future digital jobs and skills.

Environmental

- Reduces the use of paper by offering digital lessons, notes, and assignments.
- Minimizes travel to access learning centers, saving time and energy.

Overall

- Empowers rural students with modern learning tools.
- Strengthens teacher-student interaction and school management.
- Creates a sustainable system for continuous learning, even in low-resource areas.

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RESEARCH AND REFERENCES

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❖ Research Sources Used

1. **Government Reports** → National Education Policy (NEP 2020), Digital India program insights.
2. **UNESCO Reports** → Challenges in rural education and digital inclusion.
3. **Field Studies** → Research papers on offline-first learning apps and rural internet access.
4. **Existing Solutions** → Study of platforms like **Khan Academy, Rumie, BRCK, eGranary**, and their limitations.
5. **Surveys** → Statistics on rural India's device ownership, internet access, and literacy levels.

❖ Reference Links

1. **National Education Policy (NEP 2020)** –
   https://www.education.gov.in/nep
2. **Kolibri** – https://learningequality.org/kolibri/about-kolibri/
3. **DIKSHA**– https://pmevidya.education.gov.in/diksha.html
4. **BRCK Education – Kio Kit** – https://brck.com

❖ Demo Link : https://v0-digital-learning-platform-ebon.vercel.app/

Feature	Kolibri	DIKSHA	Punjab Educare	Our App
Offline Access	✅	⚠️	❌	✅
Local Language Support	✅	✅	✅	✅
Teacher Dashboard	✅	✅	⚠️	✅
Parent Communication	❌	❌	❌	✅
AI Tutor (Voice + Adaptive)	❌	❌	❌	✅
Offline Code Editor	❌	❌	❌	✅
Gamification & Rewards	✅	⚠️	❌	✅
Sync Method	✅	✅	✅	✅
Accessibility (SLD/Disabled)	❌	❌	❌	✅
Community Integration	❌	⚠️	❌	✅
Scalability & Future	✅	✅	⚠️	✅